

Infrared Burglar Alarm

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Passive infrared (PIR) sensors detect IR light radiating from objects in their field-of-view. These are most often used in PIR based motion detectors. Presented here is a circuit of a burglar alarm based on a PIR sensor. The alarm gets activated at night and deactivated in the morning. The activation and deactivation of the alarm happens automatically, without any hu-

man intervention. Author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the burglar alarm based on a PIR sensor is shown in Fig. 2. It is built around binary counter CD4060 (IC1), PIR motion-sensor module that is connected to connector CON1, tone generator IC UM3561 (IC2), transistors BC547 (T1 and T2) and SL100 (T3) and a few other components.

The PIR sensor works on the basis of detection of IR from the heat released by a human body. It has two piezoelectric sensors inside. One sensor uses background temperature to create reference and keeps it in standby mode. The second sensor detects IR from the human body. It compares it with the reference and gets activated to generate output volt-

age of about 3.3V DC.

The sensor also has an amplifier and a timer circuit. It has three pins, namely, Vcc, GND and output (O/P) for interfacing. It works on 5V DC. The output can be used to drive a relay, triac, etc for operating AC loads.

The sensor has two inbuilt presets for adjusting time delay and sensitivity (detection range). Its high output time can be set between 30 seconds and three minutes using the time-delay preset. Similarly, detection range can be set between a few centimetres to a few metres (depending on the selected sensor module) using the sensitivity preset.

The circuit uses an alarm generator, a timer and a white LED lamp, in addition to the PIR sensor module. The timer circuit is meant for activating the PIR sensor three hours after sunset.

Light-dependent resistor LDR1 is used as a light sensor for controlling the timer built around IC1. LDR1 and resistor R1 make a voltage divider network whose output is fed to reset

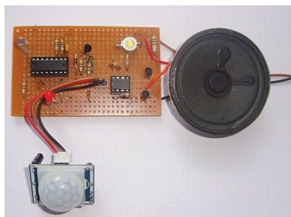


Fig. 1: Author's prototype

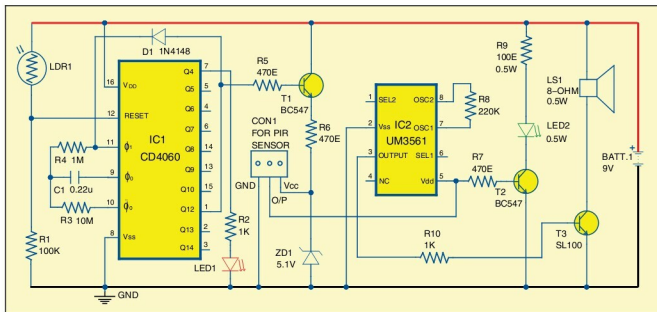


Fig. 2: Circuit diagram of the infrared burglar alarm